

MATH 114 - Computational Stochastics

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1 Overview

1.1 Introduction

Stochastic models describe real life phenomena like motion of the cell, shape of polymers stock prices, and more. What can we compute analytically and numerically from these models?

Key questions:

1. What is the distribution of the system?
2. How can we sample from the distribution?
3. How can we compute mean, variance, and higher moments of these systems?

Main topics:

1. Sampling RVs
2. Markov chain Monte Carlo (MCMC)
3. Stochastic differential equations (numerical & computation)

1.2 Sampling

- Generating "hard" RVs from "easy" RVs.
- Compute "hard" integrals.

ex Area of the unit circle, $A = \pi$

Let $U_1 \sim U(-1, 1)$ and $U_2 \sim U(-1, 1)$

Let $x = \begin{cases} 0 & \text{if } (u_1, u_2) \text{ outside circle} \\ 1 & \text{if } (u_1, u_2) \text{ inside circle} \end{cases}$

$\implies P(X = 1) = \frac{A}{4}, P(X = 0) = 1 - \frac{A}{4}$

$\mathbb{E}(X) = \frac{A}{4}$

After N trials we generate $X_1, X_2, \dots, X_n$ iid

By the strong law of large numbers

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N X_i = \frac{A}{4} = \frac{\pi}{4}$$

1.3 MCMC

Construct Markov chains to generate target distribution.

ex Ising model

2 Markov Chains

Def: A stochastic process is a family of RVs, $X_t \in S$, indexed by t , where $S \subseteq \mathbb{R}^d$ is a state space. Typically, $t \in [0, \infty)$, or $t \in [a, b]$, or $t \in \{0, 1, 2, 3, \dots\}$, etc.

- A discrete time stochastic process, is a sequence of RVs, $X_n \in S$, $n = 0, 1, 2, \dots$ or $n = 1, 2, 3, \dots$

- If S is finite or countably infinite then the stochastic process is a discrete state process.

First, we will examine discrete-time, discrete space, stochastic process.

Two basic questions:

- Find the joint distribution

$$P(X_0 = i_0, X_1 = i_1, \dots, X_n = i_n) \\ i_0, i_1, \dots, i_n \in S$$

- Find the long-time behavior of the distribution of X_n :

$$P(x_n = i), i \in S, n \gg 1$$

Def: Markov processes

- A stochastic process $\{X_n\}_{n=0}^{\infty}$ is a Markov process if

$$P(X_{n+1} = i_{n+1} | x_0 = i_0, \dots, x_n = i_n) \\ = P(X_{n+1} = i_{n+1} | X_n = i_n) \\ \forall n \geq 0, \forall i_0, \dots, i_{n+1} \in S$$

- A Markov process is time homogeneous if:

$$P(x_{n+1} = j | x_n = i) \\ = P(X_1 = j | X_0 = i) \\ \forall n \geq 0, \forall i, j \in S$$

Here we consider discrete time/state, time homogeneous Markov Chains (MC).

Def: Transition probability

$$P(i, j) = P(X_1 = j | X_0 = i) \\ = P(i \rightarrow j) \quad \forall i, j \in S$$

Def: Transition matrix

$$P = [p(i, j)] \text{ if } S = \{1, 2, 3, \dots\} \text{ or } \{0, 1, 2, \dots\} \\ \text{Here } i, j \text{ are states and also are row/column labels}$$

For any $n \geq 1, i_0, \dots, i_n \in S$, we have:

$$\begin{aligned}
 & P(X_0 = i_0, \dots, X_n = i_n) \\
 &= P(X_0 = i_0)P(X_1 = i_1|X_0 = i_0)P(X_2 = i_2|X_1 = i_1, X_0 = i_0) \\
 &\dots P(X_n = i_n|X_{n-1} = i_{n-1}, \dots, X_0 = i_0) \quad [\text{conditional probability}]
 \end{aligned}$$